

How Pitch Sequencing and Tempo Drive Swing-and-Miss Deception

A Driver Analysis of the Deception+ Whiff and Timing Residuals, 2025–2026

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Executive Summary

This analysis builds on Deception+, a from-scratch pitcher deception metric scored on two complete seasons of Statcast pitch-level data (2025–2026), by asking what season-level pitcher traits explain the residual the metric already measures: whether a pitcher gets more whiffs, and bigger contact-timing misses, than their own stuff, location, opponent, and game context alone would predict. Two scored residuals were regressed on 14 season-level driver features (pitch-to-pitch velocity and movement change, tempo, release-point and approach-angle consistency, tunnel separation, spin mirroring, repertoire size) using ordinary least squares, the same coefficient-table methodology used throughout this kind of analysis.

Key Findings:

- Driver models explain a modest but statistically real share of each residual: $R^2 = 0.047$ (whiff, $n = 1,490$) and $R^2 = 0.071$ (timing, $n = 1,457$), both significant overall ($F = 5.20$, $p < 0.0001$ and $F = 7.82$, $p < 0.0001$)
- 6 of 14 features are significant predictors of the whiff residual ($p < 0.05$); 3 of 14 for the timing residual
- Same-pitch-type repeat rate is the largest-magnitude significant predictor in *both* models (whiff $+0.0231$, $p < 0.001$; timing $+0.502$, $p < 0.0001$) and runs counter to the "always mix your pitches" convention: repeating a pitch type more often within an at-bat predicts more deception, not less
- Velocity gap from the previous pitch and vertical-approach-angle consistency across a pitcher's repertoire are validated, same-signed predictors in both models
- Deception+'s own held-out year-ahead forecast test, not a mere year-effect coefficient, confirms both outcomes studied here are temporally stable: whiff cross-validated $\Delta R^2 = +0.061$ ($p < 0.0001$), timing $\Delta R^2 = +0.102$ ($p < 0.0001$)

Research Methodology & Analytical Framework

Analytical Approach

- Deception+ scores every pitch against a per-pitch gradient-boosted expectation (stuff, plate location, batter and catcher tendency, count, park, platoon, season, in-game fatigue, times through the order, and game state), fit with 5-fold cross-validation grouped by pitcher so a pitcher's own pitches never inform their own expectation.
- The residual (actual minus expected) is empirical-Bayes shrunk toward the league mean using a Paule–Mandel between-pitcher variance estimate, then scored on a 100/10 index.
- This report regresses that shrunk residual on 14 pitcher-season traits engineered independently of

the scoring model: release-point, arm-angle and vertical-approach-angle spread across a pitcher's repertoire, velocity gap and repeat rate from the previous pitch, tunnel differential, spin-axis mirror score, pace, pitch-mix entropy, and repertoire size.

- **Complete Coefficient Analysis:** ordinary least squares with standard errors, p -values, and 95% confidence intervals for every feature in both models. (The project's existing driver analysis uses Ridge regression and Random Forest permutation importance for a predictive read; this report adds the classical significance-testing layer the coefficient-table format below needs.)
- Deliberately excluded: Savant's own swing-outcome stats (miss distance, early/late/tied-up/flailed percentages), since they are outcomes of the same swings the residual already scores, and the pitch-tempo pull's duplicate pace column, confirmed a byte-for-byte copy of the first.

Data Scope & Model Performance

- Two complete regular seasons (2025–2026), data through September 27, 2026
- 1,505 pitcher-seasons scored overall; 1,031 qualified
- **Whiff Driver Model:** $R^2 = 0.047$, Adjusted $R^2 = 0.038$, $F = 5.20$ (14 features, $df = 1,475$, $n = 1,490$)
- **Timing Driver Model:** $R^2 = 0.071$, Adjusted $R^2 = 0.062$, $F = 7.82$ (14 features, $df = 1,442$, $n = 1,457$)
- **Temporal Validation:** both outcomes pass a genuine held-out year-ahead forecast test (see Temporal Stability Analysis)

Coefficient Analysis — Whiff Residual Driver Model

Model Equation

Whiff residual = $-0.0617 + 0.0014 \times (\text{Pace between pitches}) + 0.0019 \times (\text{Velocity gap from previous pitch}) + 0.0231 \times (\text{Same-pitch-type repeat rate}) - 0.0040 \times (\text{VAA consistency across repertoire}) + 0.0023 \times (\text{Release extension}) + 0.0001 \times (\text{Spin-axis mirror score}) + \dots$ (8 more terms)

Tier 1: Statistically Significant Predictors ($p < 0.05$)

Pace between pitches (tempo_bases_empty_sec)

- Coefficient: $+0.00140$ — p -value: $< 0.0001^{***}$ — 95% CI: $[0.00091, 0.00188]$
- Working slower between pitches predicts a higher whiff residual. The residual's own standard deviation across qualified pitcher-seasons is 0.0145, so a 5-second pace difference moves the prediction by about half a residual standard deviation.

Velocity gap from previous pitch (avg_velocity_gap_from_prev)

- Coefficient: $+0.00186$ — p -value: $< 0.0001^{***}$ — 95% CI: $[0.00100, 0.00272]$
- Bigger speed changes between consecutive pitches predict more whiffs than stuff and location alone would, consistent with classic pitch-sequencing theory.

Same-pitch-type repeat rate (repeat_pct)

- Coefficient: +0.02309 — p -value: $< 0.001^{***}$ — 95% CI: [0.01017, 0.03601]
- The largest-magnitude significant predictor in the model. Counter-intuitive: throwing the same pitch type again within an at-bat more often is associated with *more* whiffs than expected, not fewer. See Cross-Model Strategic Insights.

VAA consistency across repertoire (vaa_cross_pitch_std)

- Coefficient: -0.00396 — p -value: 0.0073^{**} — 95% CI: $[-0.00685, -0.00107]$
- Less spread in vertical approach angle across a pitcher's pitch types, a more consistent tunnel, predicts a higher residual.

Release extension (release_extension_mean)

- Coefficient: +0.00229 — p -value: 0.0187^* — 95% CI: [0.00038, 0.00419]
- More extension toward the plate is associated with a modestly higher residual.

Spin-axis mirror score (spin_mirror_score_mean)

- Coefficient: -0.00007 — p -value: 0.0356^* — 95% CI: $[-0.00013, -0.00001]$
- A lower mirror score, consecutive pitches sharing a more similar or perfectly mirrored spin axis, predicts a higher residual: a real, if small, tunneling effect.

Tier 2: High-Impact Non-Significant Contributors

Season-average arm angle (arm_angle_szn_avg)

- Coefficient: +0.00004 — p -value: 0.3623 — Permutation importance: 0.00925
- The highest permutation importance of any non-significant whiff-model feature, but its OLS p -value gives no confidence in the sign.

Repertoire size (n_pitch_types)

- Coefficient: +0.00067 — p -value: 0.1798 — Permutation importance: 0.00195
- A bigger repertoire trends toward more deception credit, but not reliably.

Coefficient Analysis — Timing Residual Driver Model

Model Equation

Timing residual = $-0.4559 + 0.0643 \times (\text{Velocity gap from previous pitch}) + 0.5018 \times (\text{Same-pitch-type repeat rate}) - 0.0699 \times (\text{VAA consistency across repertoire}) + \dots$ (11 more terms)

Tier 1: Statistically Significant Predictors ($p < 0.05$)

Velocity gap from previous pitch (avg_velocity_gap_from_prev)

- Coefficient: +0.06427 — p -value: $< 0.0001^{***}$ — 95% CI: [0.05047, 0.07807]

- The single most reliable predictor in either model. Timing’s own residual standard deviation is 0.232, so this effect is proportionally similar in size to the whiff-model version of the same feature.

Same-pitch-type repeat rate (repeat_pct)

- Coefficient: +0.50183 — p -value: $< 0.0001^{***}$ — 95% CI: [0.29457, 0.70908]
- Confirms the whiff-model finding in an independent outcome: repeating a pitch type more often predicts bigger contact-timing misses, not smaller ones.

VAA consistency across repertoire (vaa_cross_pitch_std)

- Coefficient: -0.06992 — p -value: 0.0035^{**} — 95% CI: $[-0.11679, -0.02305]$
- Confirms the whiff-model finding: a more consistent approach angle across pitch types predicts more deception here too.

Tier 2: High-Impact Non-Significant Contributors

Release-point (x) consistency (release_pos_x_cross_pitch_std)

- Coefficient: -0.24432 — p -value: 0.0909 — Permutation importance: 0.00638
- The closest any non-significant timing feature comes to conventional significance, and the highest permutation importance among them.

Average vertical approach angle (vaa_mean)

- Coefficient: $+0.00665$ — p -value: 0.5419 — Permutation importance: 0.00390
- A pitcher’s average approach angle itself (as opposed to its *consistency* across pitch types, which is significant) carries real importance but no significant coefficient.

Cross-Model Strategic Insights

Statistically Validated Universal Factors

Velocity Gap From Previous Pitch

- Both Models: significant and positive (whiff $+0.00186$, $p < 0.0001$; timing $+0.06427$, $p < 0.0001$)
- Strategic Value: the single most consistently validated driver across both the rate and the magnitude dimension of swing-and-miss deception.

Same-Pitch-Type Repeat Rate

- Both Models: significant and positive (whiff $p < 0.001$, timing $p < 0.0001$)
- Strategic Value: validated in two independent scored outcomes; see Counter-Intuitive Discoveries below.

VAA Consistency Across Repertoire

- Both Models: significant and negative (whiff -0.00396 , $p = 0.0073$; timing -0.06992 , $p = 0.0035$)
- Strategic Value: pitchers whose different pitch types share a similar approach angle get more deception credit in both models, consistent with this project’s own spin-axis-gap finding in the scoring model itself.

Counter-Intuitive Statistical Discoveries

The Same-Pitch-Type Repeat Rate Paradox

- Both Models: positive, significant coefficients (whiff $p < 0.001$, timing $p < 0.0001$)
- Conventional pitching wisdom holds that varying pitch selection is what keeps hitters off balance. These two independent models instead show that repeating a pitch type more often within an at-bat predicts *more* deception, not less.
- Strategic Implication: this is correlational, not causal. A pitcher’s genuine knockout pitch is one worth repeating, and this driver regression cannot separate ”thrown again because it’s dominant” from ”dominant because it’s thrown again.” A pitch-level, within-pitcher test would be needed to go further.

Pace Does Not Generalize

- Whiff Model: significant and positive ($+0.00140$, $p < 0.0001$)
- Timing Model: not significant, and the opposite sign (-0.00366 , $p = 0.356$)
- Strategic Implication: these are two different residuals, not two readings of the same effect. A driver that matters for whether a hitter whiffs does not automatically matter for how far off their timing is when they do make contact.

Temporal Stability Analysis

Unlike a single in-sample year-effect coefficient, Deception+ already runs a genuine out-of-sample test: does a pitcher’s 2025 score predict their actual 2026 outcome rate, on top of what 2026’s own stuff-and-location expectation already explains? The test reported here is a 5-fold cross-validated R^2 delta plus a nested-model partial- F test, which cannot be mechanically guaranteed positive the way an in-sample R^2 delta is.

Whiff Model Forecast Validity

- Cross-validated ΔR^2 : $+0.0613$ — F -test p -value: < 0.0001 — In-sample ΔR^2 : $+0.0601$
- Interpretation: a pitcher’s 2025 whiff score predicts real, out-of-sample 2026 whiff rate beyond what 2026’s own expectation already explains.

Timing Model Forecast Validity

- Cross-validated ΔR^2 : $+0.1016$ — F -test p -value: < 0.0001 — In-sample ΔR^2 : $+0.0990$
- Interpretation: the strongest forecast validity of any scored component in the project.

Contrast: Two Components That Fail This Test

- Called Strike: cross-validated $\Delta R^2 = +0.0013$, $p = 0.012$ (borderline)
- Horizontal Alignment: cross-validated $\Delta R^2 = -0.0004$, $p = 0.156$ (not significant)
- This is exactly why the project's composite metric excludes both.

Strategic Implication: deception that shows up in a pitcher's 2025 whiff and timing numbers keeps showing up in their actual 2026 outcomes. That is the strongest evidence in the project that these two residuals measure a stable pitcher trait rather than noise, and a materially stronger check than an in-sample year dummy could ever provide.

Statistical Significance Hierarchy

Highly Significant Predictors ($p < 0.01$)

Whiff Model:

- Pace between pitches: $+0.00140$ ($p < 0.0001^{***}$)
- Velocity gap from previous pitch: $+0.00186$ ($p < 0.0001^{***}$)
- Same-pitch-type repeat rate: $+0.02309$ ($p < 0.001^{***}$)
- VAA consistency across repertoire: -0.00396 ($p = 0.0073^{**}$)

Timing Model:

- Velocity gap from previous pitch: $+0.06427$ ($p < 0.0001^{***}$)
- Same-pitch-type repeat rate: $+0.50183$ ($p < 0.0001^{***}$)
- VAA consistency across repertoire: -0.06992 ($p = 0.0035^{**}$)

Summary of Statistical Findings

Whiff Model:

- Total features analyzed: 14
- Statistically significant features ($p < 0.05$): 6 of 14
- Highest coefficient impact: Same-pitch-type repeat rate ($+0.0231$, $p < 0.001$)
- Most reliable predictor: Pace between pitches ($p < 0.0001$)

Timing Model:

- Total features analyzed: 14
- Statistically significant features ($p < 0.05$): 3 of 14
- Highest coefficient impact: Same-pitch-type repeat rate ($+0.502$, $p < 0.0001$)
- Most reliable predictor: Velocity gap from previous pitch ($p < 0.0001$)

Key Strategic Takeaways

1. **Sequencing Discipline, Not Mixing:** pitch-to-pitch velocity change and repeating a dominant pitch type both independently predict more deception than stuff and location alone would suggest, confirmed in two separate scored outcomes.
2. **Tunneling Consistency Is Real:** pitchers whose different pitch types share a similar vertical approach angle earn measurably more deception credit in both whiff and timing, matching this project's own spin-axis-gap finding from the underlying scoring model.
3. **Season-Level Traits Explain Little of the Residual:** R^2 of 0.047 to 0.071 means over 90% of the scored residual is still unexplained by these 14 traits. This is an explanatory layer on top of Deception+, not a predictive one, and should be read as such.
4. **Temporal Stability Is Verified, Not Assumed:** whiff and timing both pass a genuine held-out year-ahead forecast test; called strike and horizontal alignment do not, which is exactly why Deception+'s composite excludes them.
5. **Treat the Repeat-Rate Finding as Correlational:** this analysis cannot separate "thrown again because it's good" from "good because it's thrown again." A pitch-level, within-pitcher causal test is a natural next step.

Methodological Notes and Limitations

- **Small, season-level sample:** $n \approx 1,450$ – $1,490$ pitcher-seasons. Results describe between-pitcher patterns across a full season, not week-to-week variation within one.
- **Moderate multicollinearity:** pitch-mix entropy carries the highest variance inflation factor in both models ($VIF \approx 5.5$), reflecting its real mathematical relationship to repeat rate. No feature in either model exceeds the conventional $VIF = 10$ concern threshold, but the entropy and repeat-rate coefficients are best read together rather than in isolation.
- **Explanatory, not predictive:** this driver regression is a diagnostic layer only, exactly as the project's own driver-analysis module states. It does not feed back into, or change, any pitcher's scored Deception+ value.